

Computational Neuroscience — Module 04:

Cognition — Study Questions

1. What is the role of the prefrontal cortex (PFC) in cognition?
2. How does working memory relate to the approximate posterior $q(s)$ in Active Inference?
3. What is the dorsal attention network? What does it control?
4. What is the ventral attention network? How does it differ from the dorsal network?
5. How do both attention networks implement precision weighting?
6. What is the P300 event-related potential? What does it reflect?
7. When does frontal theta power increase? What does this suggest about its function?
8. Why does PFC activity decrease as a situation becomes more predictable?
9. How do the prefrontal and parietal cortex collaborate during goal-directed behavior?
10. What happens to cognition when the fronto-parietal network is disrupted?
11. How might schizophrenia be understood as aberrant precision on prediction errors?
12. How does ADHD relate to impaired precision modulation?
13. How does the computational psychiatry view of depression differ from the "chemical imbalance" model?
14. What neural mechanism might explain why anxious people overestimate threat?
15. What is the difference between top-down and bottom-up attention at the neural level?
16. How do neuromodulators (dopamine, acetylcholine, noradrenaline) interact with attention networks?
17. Can you measure free energy minimization in the brain? What would you look for?
18. How does the N400 ERP component relate to prediction error in language processing?
19. What happens in the brain when you change your mind about something?
20. Compare the computational neuroscience view of cognition with the cognitive science view (Unit 1, Module 04). What does each add?